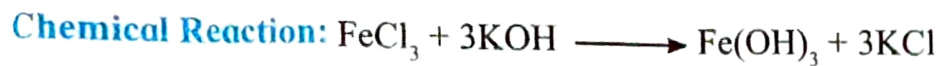


Experiment No. 13

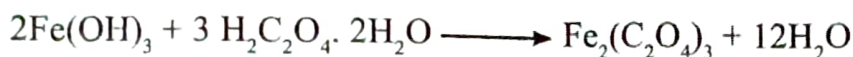
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Aim: To prepare a pure sample of the complex potassiumtrioxalatoferate(III).

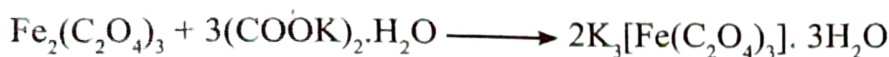
Theory: When hydrated ferric chloride is dissolved in aqueous solution of oxalic acid containing potassium hydroxide, green crystals of potassiumtrioxalatoferate(III) are obtained. The reactions involved in the formation of green crystals are as follows:



Ferric hydroxide



Ferrous oxalate



Potassium oxalate

potassiumtrioxalatoferate(III)

Apparatus: Beakers, china dish, funnel, glass rod, tripod stand, wire gauze, sand bath, burner, etc.

Chemicals: Ferric chloride, oxalic acid, potassium hydroxide, potassium oxalate.

Procedure:

1. Take 1.15 g of potassium hydroxide in a beaker add 15 mL water shake well to dissolve it.
2. Take 1 g of anhydrous ferric chloride in a another beaker add 15 mL distilled water shake well to dissolve it.
3. Transfer KOH solution in to FeCl_3 solution in small portion with constant stirring. Ferric hydroxide precipitates out. Filter the precipitates and wash it with distilled water.
4. In another beaker take 1.15 g of oxalic acid and 1.6 g of hydrated potassium oxalate and 30 mL distilled water. Warm to get clear solution.
5. Add ferric hydroxide precipitate to this above solution and stir well to dissolve it.
6. Filter the solution and transfer it to china dish, heat the china dish on sand bath to reduce its volume. Cool the concentrate to obtain fine green crystals of the potassiumtrioxalatoferate(III) complex, wash the crystals with small quantity of ethyl alcohol. Dry and weigh the crystals.
7. Shape of potassiumtrioxalatoferate(III) is octahedral.

Observation and Result:

1. Colour of potassiumtrioxalatoferate(III) is green.
2. Yield of potassiumtrioxalatoferate(III) is 3 g.
3. Shape of potassiumtrioxalatoferate(III) is octahedral.

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

1. Geometry of the complex $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$ is
 a. bipyramidal b. trigonal
 ✓ c. octahedral d. prismatic
2. Complex potassiumtrioxalatoferate(III) is
 a. ferromagnetic ✓ b. paramagnetic
 c. diamagnetic d. non-magnetic
3. The co-ordination number of iron in potassiumtrioxalatoferate(III) is
 a. three b. four c. five ✓ d. six
4. In potassiumtrioxalatoferate(III) complex, oxalate ion is
 ✓ a. bidentate b. tridentate
 c. tetradenate d. polydenate
5. The colour of potassiumtrioxalatoferate(III) complex is
 a. colourless b. yellow c. red ✓ d. green

Short answer questions

1. Write IUPAC name of the compound $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$

Ans.

Potassium trioxalatoferate (III)

2. Why does the compound potassiumtrioxalatoferate(III), does not give test for ferric ion

Ans.

Because it contains Fe (III) as complex ion $[Fe(C_2O_4)_3]^-$ and not as Fe^{3+} ions

3. Write two examples of bidentate ligands other than oxalate ion.

Ans.

i) ethylenediamine $H_2N-CH_2-CH_2-NH_2$
 ii) Acetylacetonate ion

4. Why the complex potassiumtrioxalatoferate(III) is paramagnetic in nature?

Ans.

Because in complex potassiumtrioxalatoferate, iron has unpaired electron.

5. What is meant by equimolar proportion?

Ans.

Equimolar means having the same molar concentration of solute in solvent

Remark and sign of teacher: